

Integrating artificial intelligence into Ayurveda: Pathways, potentials, and challenges

Introduction

Ayurveda, the ancient system of holistic medicine, is grounded in experiential knowledge, personalized treatment approaches, and a comprehensive view of health and disease. Its time-tested principles continue to serve as a foundation for preventive, promotive, and curative healthcare practices. In recent years, the rapid advancement of artificial intelligence (AI) has opened new avenues for all sectors of healthcare including Traditional Medicine(TM) and Ayurveda. The World Health Organization (WHO), in partnership with the International Telecommunication Union (ITU), launched the Focus Group on AI for Health (FG-AI4H) in 2018, which later led to the formation of the Global Initiative on AI for Health (GI-AI4H) in July 2023, along with ITU and WIPO, as a long-term structure to advance AI in healthcare. WHO has since published several key documents: Ethics and Governance of Artificial Intelligence for Health: WHO Guidance Executive Summary (28 June 2021), Global Strategy on Digital Health 2020–2025 (18 August 2021), Generating Evidence for Artificial Intelligence-Based Medical Devices: A Framework for Training (17 November 2021), Regulatory Considerations on Artificial Intelligence for Health (19 October 2023), and Ethics and Governance of Artificial Intelligence for Health: Guidance on Large Multi-Modal Models (25 March 2025), highlighting its ongoing commitment to ethical, strategic, and regulatory aspects of AI in health.^[1]

The Ministry of Ayush is actively advancing AI integration in Indian Systems of Medicine through curriculum development, AI-powered e-learning, Ayush Grid applications, and global expert collaborations—highlighted by the Global Experts Convene on AI in Traditional Medicine held at AIIA, New Delhi (12–13 Sept 2024).^[2] AI technologies such as machine learning, natural language processing (NLP), and data integration tools offer promising applications in clinical decision-making, educational innovations, pharmacovigilance, and the digitization of classical Ayurvedic knowledge systems.^[3,4] This appraisal aims to explore the current and potential applications of AI in Ayurveda, identify key opportunities and challenges in this integrative journey, and propose strategic directions for responsible and epistemologically aligned adoption of AI within the Ayurvedic framework.

Scope and Applications of Artificial Intelligence in Ayurveda

Artificial intelligence in traditional medicine: Real-world applications and opportunities

Globally, AI is being increasingly integrated into traditional medicine systems to support evidence generation, drug discovery, and safety monitoring. For example, in China, AI has been effectively applied in traditional Chinese medicine (TCM) through the development of machine learning models to classify syndromes, predict therapeutic outcomes, and mine historical clinical records for drug discovery.^[5] A notable instance includes the use of AI-driven knowledge graphs and text-mining algorithms to extract pharmacological insights from TCM classical texts, supporting reverse pharmacology and identifying novel therapeutic leads.^[5,6]

Adverse drug reaction and pharmacovigilance

AI plays a crucial role in enhancing drug safety by analyzing large datasets to detect patterns of adverse drug reactions (ADRs) that may not surface during clinical trials. By predicting potential risks early, AI supports safer, faster, and more informed decision-making in drug development and post-marketing surveillance. Researchers opined that AI-based tools for ADR detection from Ayurvedic hospital databases and public health forums, showcasing AI's role in proactive safety monitoring.^[7] These examples illustrate how AI applications such as text mining, clinical data analytics, and predictive modeling can significantly advance traditional medicine by enhancing therapeutic safety, optimizing drug development, and bridging classical knowledge with contemporary biomedical research.

Digitization and semantic mapping of classical texts

Ayurveda's classical literature, primarily in Sanskrit, holds immense medical knowledge developed over centuries. However, access and analysis are hindered by linguistic complexity and a lack of digitization. Emerging AI

technologies such as NLP and optical character recognition can address this challenge by converting ancient texts into machine-readable formats and enabling semantic mapping of Ayurvedic concepts across classical works.^[8] This paves the way for building ontological databases and knowledge graphs that enhance teaching, cross-textual research, and contextual understanding. Moreover, it opens doors for interdisciplinary collaboration between Ayurveda and modern biomedical sciences.

Clinical decision support systems (CDSS)

Ayurvedic practice depends on the clinician's integrative judgment, drawing from experiential knowledge and pattern recognition. AI-driven clinical decision support systems (CDSS) can enhance this process by analyzing patient data to propose likely diagnoses, suggest suitable Ayurvedic formulations, and customize treatment plans. These tools are especially valuable in settings lacking experienced practitioners or in managing complex, multi-morbid conditions. Additionally, CDSS can evaluate risks such as herb-drug interactions and contraindications in real time, thereby improving patient safety and supporting evidence-based decision-making. Crucially, CDSS should serve as an aid, not a replacement, for the Vaidya's clinical expertise.^[9]

Ayurvedic research and big data integration

The current landscape of Ayurvedic research is witnessing a gradual shift from fragmented, narrative-based documentation to structured, data-driven approaches, largely propelled by national digital initiatives such as Ayush Health Management Information System (AHMIS), Ayush Grid, and the NAMASTE portal. These platforms have laid the groundwork for standardizing clinical records, terminologies, and diagnostics across institutions, enabling the creation of interoperable datasets necessary for AI applications. The establishment of the *Ayurinformatics* in Ayurveda institutes further exemplifies this transformation—serving as a pioneering center for computational drug discovery, molecular docking, and reverse pharmacology using AI tools. With these foundations, the future scope of AI in Ayurveda lies in developing predictive models for treatment responses, integrating *Prakriti*-based classifications with genomics and metabolomics, and enabling real-time analytics for pharmacovigilance and public health decision-making. As big data ecosystems mature and interdisciplinary collaborations strengthen, AI is poised to redefine Ayurvedic research by bringing reproducibility, scalability, and precision to traditional knowledge systems.

Challenges and Cautions

Epistemological conflicts

One of the primary challenges in integrating AI with Ayurveda lies in their fundamentally different epistemological frameworks. Ayurveda is built upon holistic causality, qualitative analysis, and experiential understanding, emphasizing individualized treatment and the practitioner's intuition (*Yukti*). In contrast, AI relies on statistical inference, pattern recognition, and quantifiable parameters to conclude. Translating nuanced Ayurvedic concepts such as *Rasa* (taste), *Guna* (Properties), *Vipaka* (post-digestive effect), *Virya* (potency), *Prabhava* (specific action), and *Agni* (digestive/metabolic fire) into discrete, measurable variables is inherently difficult. This runs the risk of oversimplifying or misrepresenting complex traditional constructs, potentially diluting the depth and context of Ayurvedic wisdom. Thus, efforts to incorporate AI must be rooted in a deep understanding of Ayurvedic philosophy to ensure epistemological alignment and contextual relevance.

Data quality and standardization

Effective AI implementation in Ayurveda depends on access to high-quality, standardized clinical data—something historically hindered by inconsistent documentation, non-uniform diagnostic protocols, and diverse terminologies. Addressing this gap, national initiatives like Ayush Grid and the AHMIS have been launched to digitize and streamline patient records across Ayush institutions. AHMIS, enables structured data collection, real-time analytics, and standardized reporting formats. These efforts are gradually creating interoperable, anonymized datasets crucial for training AI models and developing clinical decision support systems. Integration with the National Digital Health Mission further ensures compatibility with broader health data frameworks. While implementation challenges remain, these platforms form a foundational ecosystem for scalable, evidence-based AI applications in Ayurveda.

Ethical and legal issues

Using AI in Ayurveda brings up important ethical and legal issues, especially around patient privacy, how AI makes decisions, and who is responsible if something goes wrong. Unlike modern medicine, Ayurveda does not yet have clear rules for using digital tools and AI. There are still unanswered questions about who owns the data, how to get proper patient consent, and who is legally responsible if an AI-related mistake happens. There's also a concern that

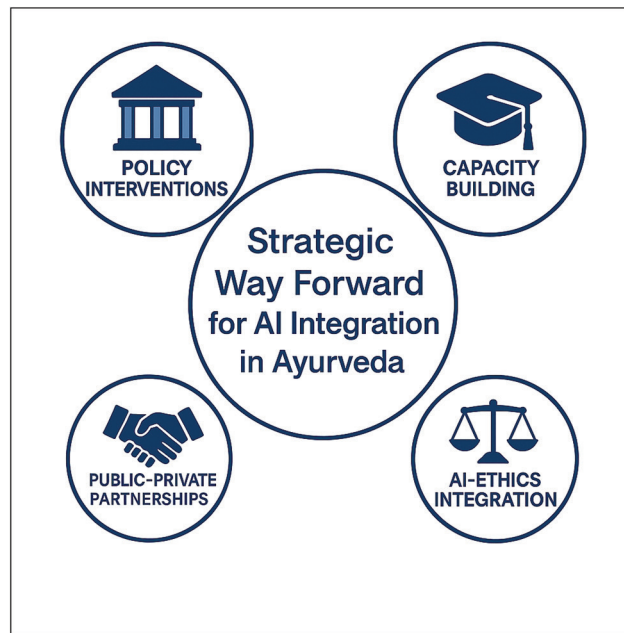


Figure 1: Strategic way forward for artificial intelligence integration in Ayurveda

too much trust in AI could reduce the role of *Yukti*—the skill, judgment, and experience of the Ayurvedic doctor. AI should support, not replace, the expert thinking of the physician. To use AI safely and wisely in Ayurveda, proper ethical guidelines and clear rules must be created.

Resource and skill gaps

One of the major barriers to AI adoption in Ayurveda is the limited digital and interdisciplinary literacy among Ayurvedic professionals. Most *Vaidyas* and researchers lack formal exposure to AI tools, data science, and computational methods, which restricts both the development and effective use of such technologies. Moreover, there is minimal collaboration between Ayurvedic institutions and AI or tech experts, creating a disconnect between traditional knowledge and modern innovation. To bridge this gap, focused investments are needed in digital infrastructure, capacity-building programs, and collaborative platforms that unite Ayurvedic scholars with data scientists, developers, and policy-makers. Such efforts are essential to position Ayurveda within the rapidly advancing domain of intelligent healthcare.

Strategic way forward

To make the best use of AI in Ayurveda, a thoughtful and well-rounded approach is needed. In addition to the present initiatives, Government of India should take lead at Central level to creating guidelines in parlance with WHO, for how AI can be safely used, how digital data is managed, and how AI tools should be tested in Ayurvedic settings. At the same time, intensive training programs should include AI and digital health topics in postgraduate and PhD courses so that future Ayurvedic professionals are prepared for this integration. Partnerships with AI companies, startups, Ayush /Health Sciences Universities and academic institutions should be encouraged to develop India-specific AI tools that align with Ayurvedic principles. For this, it is important to create secure, shared clinical data systems and adopt common terms and formats for Ayurvedic diagnosis and treatments. Lastly, AI ethics should be based on Indian cultural values, ensuring that the use of AI supports personalized, safe, and compassionate patient care [Figure 1].

Conclusion

The integration of AI in Ayurveda offers a transformative opportunity to enhance its relevance, reach, and scientific foundation. To realize this potential, AI must be applied with epistemological sensitivity and ethical care. A strategic approach—encompassing policy, education, data systems, and culturally aligned ethics—can help Ayurveda evolve into a future-ready, globally impactful healthcare system without losing its traditional essence.

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Conflicts of interest

There are no conflicts of interest.

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